“Clothing Brand Web System for Online Retail Operations”

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Approval Sheet

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SOFTWARE DESIGN PROJECT INFORMATION

2nd Semester, SY 2025-2026

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Project Title	Clothing Brand Web System for Online Retail Operations
Project Concentration Area	Web Systems, E-Commerce Technologies, Database-Driven Applications
Design Objectives	The general objective is to provide a comprehensive design and development plan for the clothing brand web system. This includes the features and website functions that will be implemented in the project. Specifically, it aims to: 1. Develop a Web Based system that; • That showcase the clothing (Hoodies, Polo, Pant etc..) of Doing Good • Shows the categories of clothe based gender • Has a database management for products, users, orders, and inventory 2. Test and evaluate the system's accuracy and performance
Constraints	
Performance Constraints (Response Time)	Performance constraint relates to the speed and responsiveness of the website, measured by load and response time. Poor performance reduces user satisfaction. Therefore, faster systems are more effective. Hence, the design with the best performance is considered the best design.

Security Constraints (Data Protection Level)	Security constraint relates to protecting user data and system integrity through secure methods. Weak security increases the risk of data breaches. Therefore, stronger security is more reliable. Hence, the design with the highest security is considered the best design.
Scalability Constraints (Number of Concurrent Users)	Scalability constraint relates to the system's ability to handle growth in users and data. Poor scalability leads to slowdowns or failures. Therefore, scalable systems are more reliable. Hence, the design with the highest scalability is considered the best design.
Efficiency (Inference Time)	Efficiency constraint relates to the optimal use of resources such as time, cost, and system power. Inefficiency increases costs and slows performance. Therefore, efficient systems are more effective. Hence, the design with the highest efficiency is considered the best design.
Technology Constraints (Technology Stack)	Technology constraint relates to the tools and platforms used in development. Poor technology choices can limit performance and maintenance. Therefore, modern and compatible technologies are more effective. Hence, the design with the best technology is considered the best design.
Other constraints: These constraints do not affect each design; therefore, these were not included in selecting the best design.	
Public Health	Websites contribute to public health by sharing reliable information, promoting safe online practices, and protecting user data privacy. They can also support awareness campaigns and provide access to important services.
Welfare	Websites improve user welfare by offering convenient access to products and services, secure transactions, and efficient customer support, making everyday activities easier and more accessible.
Social	Websites strengthen social connections by enabling communication, interaction, and community building through online platforms, feedback systems, and shared content.
Global	Websites allow businesses and organizations to reach international audiences, operate across different regions, and serve users worldwide without geographical limitations.
Cultural	Websites can respect and represent different cultures by providing multilingual support, inclusive design, and content that considers cultural values, traditions, and diversity

Standards	
Standard A - WCAG 2.1 Level AA	This standard makes sure websites are accessible to people. In the project, it was used to improve text contrast, add text to product images, and ensure the site can be used with a keyboard.
Standard B - OWASP Top 10	This standard lists the most common security risks for web systems. In the project, it helped guide secure login, protect user data, and prevent issues like unsafe input or weak passwords.

Abstract

This study presents the design and development of a Clothing Brand Web System for Online Retail Operations aimed at improving the digital presence and operational efficiency of a clothing business. The system provides an interactive platform where customers can browse products, create accounts, and place orders, while administrators can manage product listings, inventory, and transactions through a centralized interface. It follows a three-tier architecture consisting of a front-end, back-end, and database to ensure organized data processing and system functionality.

The system was developed using modern web technologies and evaluated based on performance, security, scalability, and usability. Testing results show that the system performs efficiently with fast response time, reliable data handling, and secure user authentication. Additionally, recognized standards such as accessibility and security guidelines were applied to enhance user experience and protect sensitive information. Overall, the system offers a scalable, secure, and efficient solution that supports the growth and competitiveness of the clothing brand in the online market.

Keywords: Web System, Online Retail, E-commerce, Clothing Brand, Inventory Management, Order Processing, User Interface, Database Management, Web Development, System Performance, Security, Scalability.

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List of abbreviation

API – Application Programming Interface
CSS – Cascading Style Sheets
HTML – HyperText Markup Language
JS – JavaScript
JWT – JSON Web Token
MCDM – Multi-Criteria Decision Making
OWASP – Open Web Application Security Project
PHP – Hypertext Preprocessor
SDG – Sustainable Development Goal
SQL – Structured Query Language
UI – User Interface
UX – User Experience
WCAG – Web Content Accessibility Guidelines

Definition of terms

Authentication – The process of verifying the identity of a user before granting access to the system.
Authorization – The process of determining what actions a verified user is allowed to perform within the system.
Backend – The server-side part of the system responsible for processing data, managing logic, and handling requests.
Database – A structured collection of data used to store information such as users, products, and transactions.
E-commerce – The buying and selling of goods or services over the internet.
Frontend – The user-facing part of the system that allows interaction through web pages and interfaces.
Inventory Management – The process of tracking and managing stock levels of products in the system.
Normalization (MCDM) – A method used to convert different values into a common scale for comparison in decision-making.
Page Load Time – The time it takes for a web page to fully display its content to the user.
Scalability – The ability of the system to handle an increasing number of users or data without affecting performance.
Security – Measures implemented to protect system data from unauthorized access or breaches.
System Architecture – The overall structure of the system, including how components interact with each other.
Three-tier Architecture – A system design divided into presentation, application, and data layers.
Web Application – A software application that runs on a web browser and is accessed through the internet.

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CHAPTER 1: THE PROJECT AND ITS BACKGROUND

This chapter highlights the need for a web system to help the clothing brand improve visibility, manage inventory efficiently, and increase sales. By applying engineering standards and a structured design process, the project ensures the system is secure, reliable, and aligned with the client's goals.

1.1 The Problem

The Doing Good Clothing business faces several challenges that limit its growth and market reach, particularly in the absence of a dedicated online platform. One of the primary concerns is the inability to expand the brand name globally, as the business currently relies on limited channels that restrict its visibility to a wider audience. This lack of digital presence prevents the business from competing effectively in the growing online clothing market.

Low sales have also been observed due to insufficient online visibility, making it difficult for potential customers to discover the products and services offered. Without a centralized platform, customers are unable to access important information about the business at any time, which affects convenience and reduces engagement. This limitation can result in missed opportunities for both customer acquisition and retention.

Furthermore, the absence of an efficient system to showcase products, prices, and updates creates a gap in communication between the business and its customers. Traditional methods of promotion are often limited and inconsistent, making it harder to maintain customer interest and trust. Therefore, there is a need for a well-designed website that can provide accessible, organized, and up-to-date information.

With the advancement of web technologies and e-commerce platforms, developing a website presents a practical solution to these challenges. A dedicated online system can enhance visibility, improve customer experience, and support business growth by enabling global reach and efficient product presentation.

1.2 The Client

The client for this project is Mr. Dexter Luis Vega, the owner of an established clothing brand that is currently operating without its own online retail system. To support the growth and digital expansion of his business, our team will be developing a web-based platform that aligns with the brand's operational needs and customer expectations.

Through the interview and data-gathering process, our team will identify the specific requirements of Mr. Vega's clothing brand. The planned system will provide structured product displays, real-time inventory management, secure customer transactions, and an organized order processing workflow. By creating this platform, the project will contribute to enhancing the brand's online presence, accessibility, and overall retail efficiency.

Table 1-1 presents the main client and engineering requirements for the Doing Good Clothing website system. The client requirements highlight the business needs, such as enabling customers to easily browse products, ensuring secure handling of user accounts and transactions, and maintaining accurate inventory tracking. These reflect the operational challenges of improving customer experience, increasing online visibility, and managing product information efficiently.
Each engineering requirement is designed to be specific and actionable, translating these business needs into technical solutions. This includes implementing a responsive and accessible website design, applying secure authentication and data protection practices, and developing a database-driven inventory system supported by backend technologies such as Node.js. These considerations ensure that the website is user-friendly, secure, and capable of supporting business growth and efficient operations.

Table 1-1. Client and Engineering Requirements / Considerations

Client Requirements / Considerations	Engineering Requirements / Considerations
The website can allow customers to browse and view clothing products easily.	The website should have a responsive design, fast loading, organized navigation, and comply with WCAG 2.1 Level AA to ensure accessible browsing.
The website can securely store user accounts and handle online transactions safely.	Implement secure authentication, encrypted data storage, and OWASP Top 10 security practices.
The website can track inventory and update stock levels automatically when orders are placed.	Create a database with automated inventory management logic using backend APIs in Node.js.

1.3 The Project

This project focuses on creating a web system for Mr. Dexter Luis Vega's clothing brand to help the business operate online. The system will allow customers to browse products and make purchases, while administrators can manage products, inventory, and orders efficiently. The system will have a front-end for users to interact with, a back end to process orders and update data, and a database to store product, user, and sales information. It will also follow accessibility and security standards to ensure it is safe and easy to use. By developing this system, our team will help the brand expand online, improve operations, and prepare for future growth.

1.4 Project Objectives

The primary objective of this project is to design and develop a comprehensive website to facilitate the online retail operations of Mr. Dexter Luis Vega's clothing brand. The system will integrate functionalities for efficient management of product catalogs, inventory control, order processing, and user interaction.

Specifically, the project aims to:

1. Develop software modules encompassing both front-end and back-end functionalities, including the following:

- dynamic product display
- secure transaction handling
- database management
- real-time inventory synchronization

2. Test and evaluate

- the system's accuracy
- performance
- security
- reliability

1.5 Scope and Delimitations

This project focuses on designing and developing a web system for Mr. Dexter Luis Vega's clothing brand to support online sales and business operations. The system will include features for managing product listings, tracking inventory, processing customer orders, and handling user accounts for both customers and administrators. The system will follow important standards to ensure it is accessible and secure.

The web system will be built using React for the user interface and Node.js for the server side, along with a database to store product and customer data. The system will be tested in controlled environments to check its accuracy and performance under normal use. This project covers only the web application and does not include developing a mobile app or connecting with external delivery or payment services beyond basic online transactions. It will support up to 1,000 users at the same time but will not include advanced performance tuning for larger traffic. Security will protect against common threats, but highly advanced cyberattacks are beyond the project's scope. Marketing, sourcing of products, and physical inventory handling outside the system are not part of this project. The user interface will focus on core functions without extensive design customization or branding details.

1.6 Design Constraints

Several constraints were considered when planning the project, and these constraints limit the system's technological features while also aiding in the refinement of design possibilities during development. However, only the top five limitations were applied in the project's development process because the remaining constraints had no direct impact on the project.

Performance Constraints (Response Time)

Performance constraint relates to the speed and responsiveness of the website, measured by load and response time. Poor performance reduces user satisfaction. Therefore, faster systems are more effective. Hence, the design with the best performance is considered the best design.

Security Constraints (Data Protection Level)

Security constraint relates to protecting user data and system integrity through secure methods. Weak security increases the risk of data breaches. Therefore, stronger security is more reliable. Hence, the design with the highest security is considered the best design.

Scalability Constraints (Number of Concurrent Users)

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Other constraints: These constraints do not affect each design; therefore, these were not included in selecting the best design.

Public Health

Websites contribute to public health by sharing reliable information, promoting safe online practices, and protecting user data privacy. They can also support awareness campaigns and provide access to important services.

Welfare

Websites improve user welfare by offering convenient access to products and services, secure transactions, and efficient customer support, making everyday activities easier and more accessible.

Social

Websites strengthen social connections by enabling communication, interaction, and community building through online platforms, feedback systems, and shared content.

Global

Websites allow businesses and organizations to reach international audiences, operate across different regions, and serve users worldwide without geographical limitations.

Cultural

Websites can respect and represent different cultures by providing multilingual support, inclusive design, and content that considers cultural values, traditions, and diversity

1.7 Engineering Standards

The engineering standards used in this project ensure that the clothing brand web system is accessible, secure, reliable, and aligned with accepted practices for modern web applications. These standards support the system's design goals and help maintain quality throughout development.

WCAG 2.1 Level AA (Web Accessibility Standard)

This standard ensures that the website is accessible to all users. It supports the project's goal of creating an easy-to-use interface by guiding text contrast, keyboard navigation, readable layouts, and alt-text for product images. This directly strengthens the system's usability requirement.

OWASP Top 10 (Web Security Standard)

This standard lists the most common web security risks. It connects to the project's constraint on data protection by helping identify and prevent issues such as weak authentication, unsafe input handling, and data exposure. This ensures that user accounts, transactions, and stored data remain secure.

ISO/IEC 25010 (Software Quality Standard)

This standard guides how software quality is evaluated. It supports the project's testing requirements by defining expectations for performance, reliability, usability, and security. This connects to your objective of ensuring that "all functions operate correctly" and the system responds efficiently.

ISO/IEC 27001 (Information Security Standard)

This standard provides principles for protecting sensitive information. It supports the project's need for secure data storage and processing, especially for customer information, orders, and administrator accounts. It directly connects to your security constraints and database requirements.

ISO/IEC 14543-3 (System Architecture Standard)

Although originally intended for electronic systems, this standard is used here to support a clear and organized system architecture. It helps guide how system components communicate and work together, which connects to your objective of designing the overall structure of the system (front-end, back-end, and database).

1.8 Engineering Design Process

The image shows two people wearing stylish and comfortable clothes. Just like how clothing is planned and created, the Engineering Design Process also follows steps to make a good final product. It starts with finding a problem, thinking of ideas, testing them, and improving the design. This process helps make sure the result is useful, creative, and fits what people need.

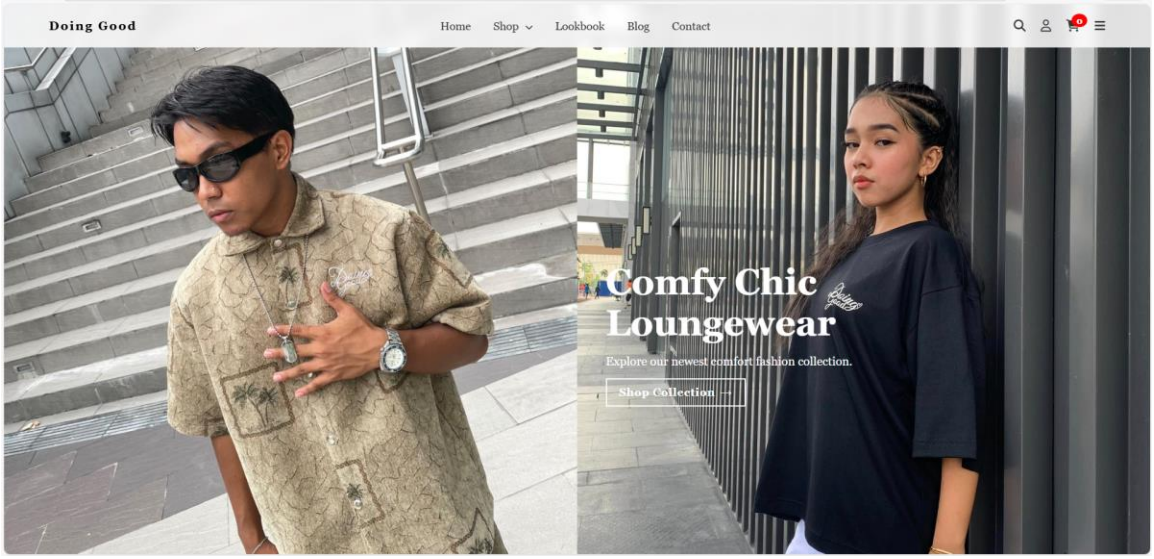

Figure 1.1 The Engineering Design Process (Teach Engineering, 2025)

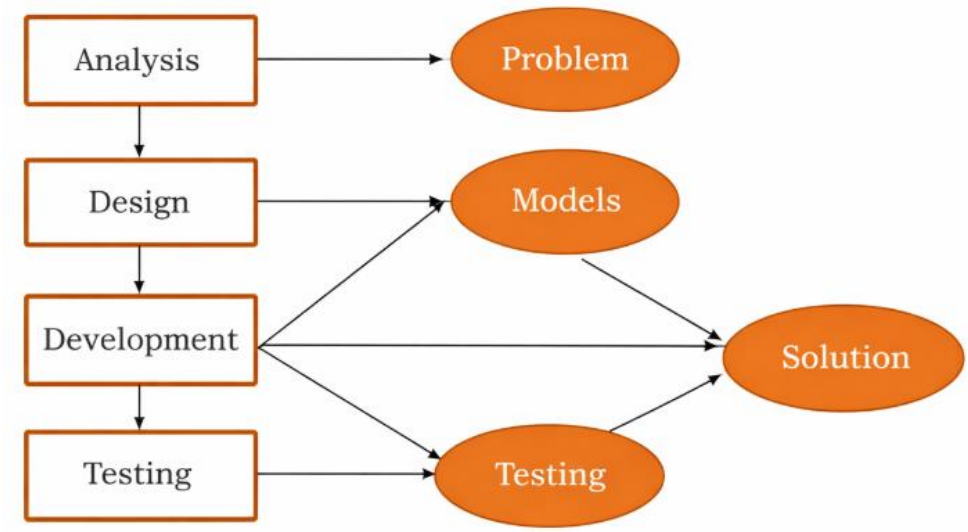

Figure 1.2 The Engineering Design Process (Teach Engineering, 2025)

The picture depicts two individuals dressed comfortably and stylishly. The Engineering Design Process follows steps to produce a high-quality final product, just like clothing is planned and made. These steps are clearly depicted in the diagram. Analysis is the first step, where we comprehend the issue or requirement. After that, we proceed to the Design stage, where concepts and models are developed. These concepts are then transformed into actual products during the development stage. Lastly, testing verifies that everything functions properly. These procedures aid in enhancing the design and guarantee that the finished product is practical, imaginative, and meets people's needs.

1.8.1 Ask: Identifying the Need and Constraints

This step entails figuring out what the client needs and comprehending the main issue. A fully functional clothing brand web system that facilitates online retail operations is required for this project. Constraints like performance limitations, security needs, financial limitations, and accessibility requirements are identified by the team. By posing crucial queries, such as "What should the system do?" Who is going to use it? What limits must it follow? the group creates a strong basis for the design.

1.8.2 Research the Problem

The team collects pertinent data and examines current systems after determining the need. This entails examining industry standards like WCAG 2.1 and OWASP Top 10, evaluating existing e-commerce platforms, and conducting client interviews to gain a deeper understanding of operational requirements. Technologies like React, Node.js, and database structures are also covered in research. This stage guarantees that the project is based on well-informed choices rather than conjecture.

1.8.3 Imagine: Develop Possible Solution

The group imagines and brainstorms different solutions in this step. This could include different database designs, inventory tracking strategies, and user interface layouts for the web system. Here, the team is encouraged to be creative and can freely explore ideas without worrying about constraints right away. To determine the most promising system structure, a number of design options are drafted.

1.8.4 Plan: Select a Promising Solution

The team brainstormed several methods the company can use to help customers explore and purchase clothing through an effective clothing business website. Since the business already has images available from the client's Facebook page, these can be used as visual content for the website.

Option A: A gallery-style website that highlights "Fashion Inspiration" photos taken from the client's Facebook page. This approach showcases different outfits and clothing items in a visually appealing way to inspire customers and attract their attention.

Option B: A product-focused website that provides detailed information for each clothing item, such as available sizes, color options, price, and care instructions, helping customers make informed purchasing decisions.

Option C: A combination approach that uses images from the Facebook page while organizing the products into clear categories such as Tops, and Bottoms, this allows customers to both view attractive clothing photos and easily navigate through the product categories.

1.8.5 Create: Build a Prototype

A working model or preliminary version of the web system is created during this stage. Simple versions of the front-end and back-end interactions, the basic UI framework, or the initial database setup are examples of prototypes. The objective is to develop a working prototype that shows how the finished system will function, not to achieve perfection. Early on in the development process, this prototype aids in identifying possible problems.

1.8.6 Test and Evaluate the Prototype

Once the prototype is created, it undergoes performance, usability, and security testing. The team examines whether the system meets the client's requirements and satisfies constraints such as response time, accessibility, and data protection. User feedback and testing tools help uncover errors, weaknesses, and areas for improvement. Evaluation ensures that the design functions reliably before moving to full development.

1.8.7 Improve: Redesign as Needed

Based on test results, the team revises and enhances the design. Improvements may involve fixing bugs, optimizing performance, strengthening security features, or reorganizing the interface for better user experience. This step reflects the iterative nature of engineering: the design is refined repeatedly until it meets the required standards and delivers a high-quality final pro

CHAPTER 2: SOFTWARE DESIGN

This chapter presents the software design of the proposed Clothing Brand Web System for Online Retail Operations. It outlines the design solution, system architecture, data management approach, applied engineering principles, and the design alternatives evaluated prior to selecting the final implementation. This chapter aims to clearly describe the system structure and demonstrate how the design meets the project objectives and constraints defined in Chapter 1.

2.1 Description of the Design Solution

The proposed design solution is a web-based e-commerce system developed to support the online retail operations of the Doing Good clothing brand. The system is designed to provide a user-friendly and efficient platform where customers can browse products, create accounts, and place orders, while administrators can manage product listings, monitor inventory, and process transactions in real time.

The system follows a three-tier architecture composed of the presentation layer, application layer, and data layer. The presentation layer (frontend) is built using React and is responsible for displaying the user interface and handling user interactions. The application layer (backend), developed using Node.js and Express.js, manages business logic, user authentication, order processing, and communication between the frontend and the database. The data layer consists of a relational database that stores essential information such as user accounts, product details, inventory records, and transaction data.

This design ensures modularity, scalability, and maintainability, allowing the system to handle multiple users while maintaining performance and reliability. It also integrates essential features such as secure login, real-time inventory updates, and organized product categorization to enhance user experience. By implementing modern web technologies and following established standards, the proposed system provides a practical, secure, and efficient solution tailored to the needs of the client's clothing business.

2.1.1 General Description

The proposed system is a web-based e-commerce platform designed to facilitate the online retail operations of the Doing Good clothing brand. It enables customers to browse products, register accounts, and place orders conveniently through a web interface, while administrators are provided with tools to manage products, inventory, and customer transactions efficiently.

The system is structured using a three-tier architecture. The presentation layer (frontend) handles user interaction through a responsive and user-friendly interface. The application layer (backend) processes request, manages business logic, and ensures secure communication between components. The data layer (database) stores and organizes all system data, including products, users, orders, and inventory records.

This structure promotes a well-organized system design that improves performance, simplifies maintenance, and supports future scalability.

The system follows a three-tier architecture:

1. **Presentation Layer (Front-end)** – built using React, responsible for user interaction and interface rendering.

2. **Application Layer (Back-end)** – developed using Node.js and Express, responsible for business logic, authentication, order processing, and inventory updates.

3. **Data Layer (Database)** – stores product information, user accounts, orders, and inventory records.

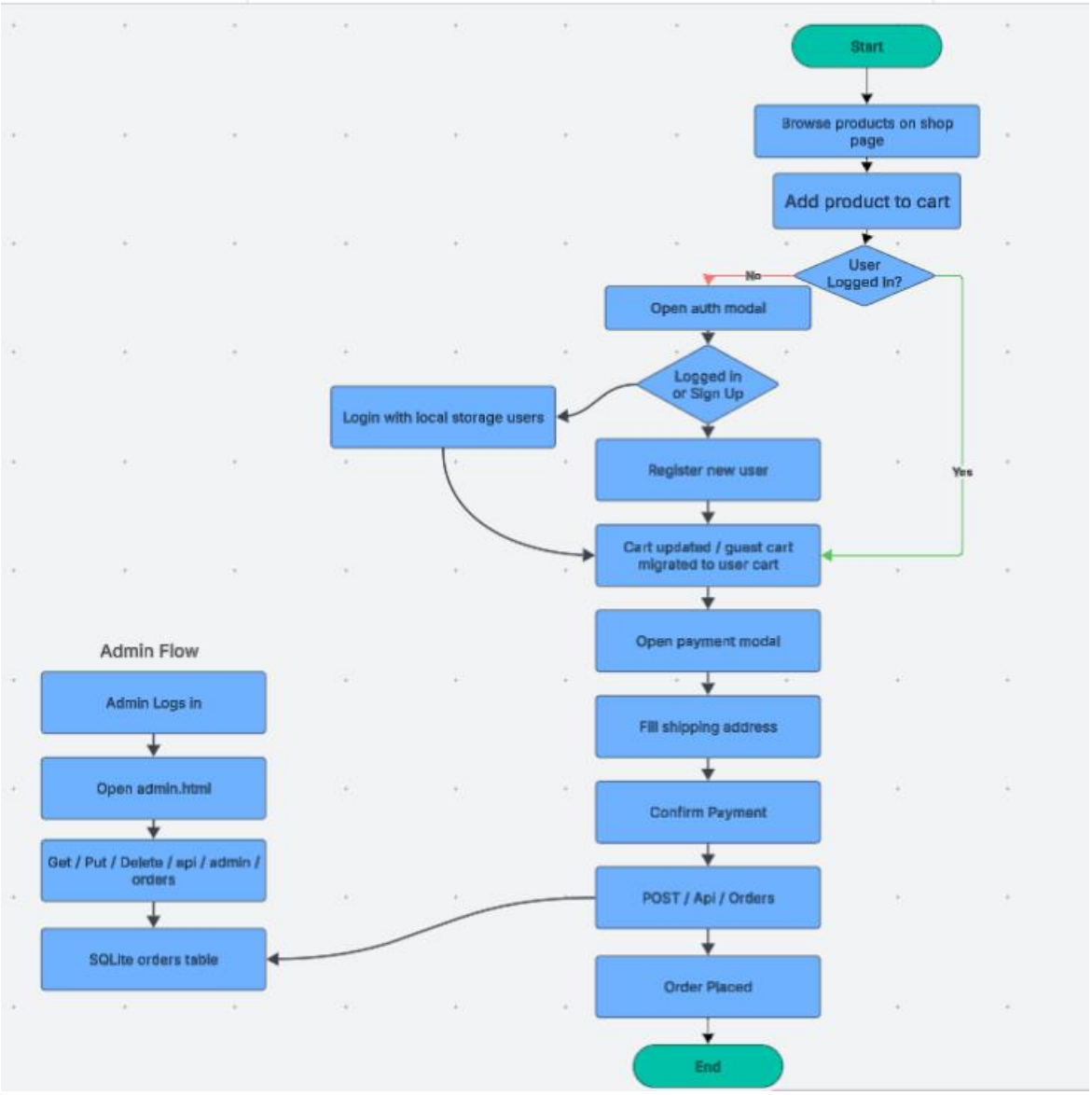

Figure 2.1 Illustrative Diagram of the System

Users access the system through a web browser, where requests are handled by the React-based front end and forwarded to the Node.js API server. The server processes each request, retrieves or updates data from the database, and returns the appropriate response to the front end for display's Administrators follow the same process but are granted access to additional system management modules.

This architecture promotes separation of concerns, simplifies system maintenance, supports scalability, and enhances overall security.

2.1.2 Engineering Principles Involved

The system design is guided by fundamental software engineering principles to ensure reliability, efficiency, and scalability:

- **Modularity** – The system is divided into independent modules such as user authentication, product management, order processing, and inventory control. This allows easier debugging, testing, and future enhancements.

- **Abstraction** – Complex processes such as database operations and security validation are handled through APIs and backend services, allowing developers to work with simplified interfaces.

- **Separation of Concerns** – The system separates the frontend, backend, and database responsibilities, reducing complexity and improving maintainability.

- **Scalability** – The architecture supports system growth by allowing the addition of more users, data, and features without significantly affecting performance.

- **Reliability and Fault Tolerance** – Error handling, input validation, and system logging are implemented to ensure stable system operation and quick issue resolution.

- **Security by Design** – Security measures such as encrypted passwords, secure authentication, and protected API endpoints are integrated from the beginning of the design process.

These principles ensure that the system remains efficient, secure, and adaptable to future requirements.

2.1.3 Prior Art Analysis

The design of the proposed system is influenced by existing e-commerce platforms that demonstrate effective online retail functionalities. These platforms provide benchmarks for product presentation, navigation, and transaction processing.

Well-known platforms such as the Nike website, Zara website, Brandy Melville website, and Urban Outfitters website offer features like organized product catalogs, user account management,

and order tracking. These systems are recognized for their responsive interfaces and efficient workflows; however, they often require significant resources and complex infrastructures.

In comparison, the proposed Doing Good web system adopts similar core features such as product categorization, user account handling, and order management, while maintaining a simplified and cost-effective architecture suitable for small to medium-sized businesses. Additionally, the system integrates advanced search functionality and brand customization, which are not consistently available across all referenced platforms.

The prior art analysis highlights that while existing systems provide strong design references, the proposed solution offers a more balanced approach by combining essential features, usability, and affordability, making it more practical for the client's operational needs.

Narrative Comparison:

The Nike website provides a highly responsive and visually structured product catalog that enhances user experience and supports quick product discovery. It also features advanced filtering options, real-time inventory visibility, and an intuitive checkout process that minimizes user effort.

In addition, the platform integrates secure authentication mechanisms and order tracking functionalities, allowing customers to monitor their purchases seamlessly. However, such enterprise level systems typically require significant development resources and infrastructure, making them less practical for small to medium-sized businesses.

In contrast, the proposed system adopts similar functional concepts and usability principles while maintaining a simplified architecture and lower implementation cost, making it more suitable for the client's operational scale and budget constraints.

Table 2.1 Prior Art Analysis Matrix

Design	Features				
	Product Catalog Organization	**Advance Search & Filtering**	**User Account Management**	**Order Tracking**	**Brand Customization**
Nike Website	x	x	x	x	
Zara Website	x		x	x	
Brandy Melville Website	x			x	
Urban Outfitters Website	x		x	x	
Generic E-commerce Template	x		x	x	
Proposed System (Doing Good)	x	x	x	x	x

Table 2.1 presents the Prior Art Analysis Matrix for the Doing Good Clothing website, comparing existing e-commerce platforms with the proposed system based on key features such as product catalog organization, advanced search and filtering, user account management, order tracking, and brand customization, where each check mark (x) indicates the presence of a feature.

While established platforms like Nike and Zara include several core features, and others such as Brandy Melville, Urban Outfitters, and generic e-commerce templates provide only partial functionality, these systems often lack a complete integration of all essential and advanced features.

In contrast, the proposed Doing Good Clothing website stands out by incorporating all key functionalities, including advanced search capabilities and full brand customization, which are not consistently present in the compared systems. This comprehensive approach ensures a more personalized, user-friendly, and efficient shopping experience. Overall, the table highlights the limitations of existing platforms and emphasizes the proposed system as a more innovative and competitive solution in the e-commerce space.

2.2 General System Architecture

The general system architecture of the proposed Clothing Brand Web System is designed to ensure efficient data processing, scalability, and reliable performance. The system follows a three-tier architecture, which separates the application into three main layers: the presentation layer, application layer, and data layer. This structure allows better organization, easier maintenance, and improved system flexibility.

In this architecture, users interact with the system through a web browser. Requests from the user are sent to the frontend interface, which communicates with the backend server through API calls. The backend processes these requests, performs necessary operations, and interacts with the database to retrieve or store data. The processed information is then sent back to the frontend for display to the user.

This approach ensures a clear separation of responsibilities among system components, enhancing system performance, security, and scalability.

2.2.1 Hardware Elements

The system does not require specialized hardware components since it is a web-based application. It operates using commonly available devices such as desktop computers, laptops, tablets, and smartphones with internet access and web browsers.

On the server side, the system can be deployed on standard web hosting platforms or cloud-based environments. This eliminates the need for expensive hardware infrastructure while ensuring accessibility and ease of deployment. The use of existing hardware resources makes the system cost-effective and practical for small to medium-sized businesses.

2.2.2 Software Elements

A. Embedded Software

Web-Based System, no embedded software is required for this project, as the Doing Good Clothing system is purely web-based and does not rely on hardware components or physical devices. Unlike systems that require microcontrollers or embedded processors, this platform operates entirely through internet browsers and server-side infrastructure. All functionalities, including data processing, user interaction, and system operations, are handled through web technologies, making the system more accessible, scalable, and easier to maintain.

B. Application Software

- **React, HTML, CSS, and JavaScript**

The frontend of the system is developed using React along with core web technologies such as HTML, CSS, and JavaScript. React enables the creation of dynamic and responsive user interfaces, allowing users to browse products, manage accounts, and interact with the website efficiently. These

technologies ensure a smooth and visually appealing user experience across different devices and screen sizes.

- **Node.js and Express.js**

The backend of the system is implemented using Node.js and Express.js, which handle server-side logic, API integration, and communication between the frontend and database. These technologies enable efficient handling of user requests, order processing, and system operations, ensuring fast and reliable performance of the web application.

- **MySQL / PostgreSQL**

A relational database such as MySQL or PostgreSQL is used to store and manage system data, including user accounts, product information, orders, and inventory. This ensures data consistency, structured storage, and efficient retrieval of information necessary for system functionality.

- **JWT and bcrypt**

Security within the system is implemented using JSON Web Tokens (JWT) for authentication and bcrypt for password hashing. JWT ensures secure user sessions and access control, while bcrypt protects user passwords by encrypting them before storage. These tools are essential in maintaining data privacy and system security.

Both customer and administrator interfaces are browser-based, eliminating the need for additional software installation and allowing easy access to the system.

C. Key Algorithms Used

- User Authentication and Authorization Algorithms

The system uses authentication and authorization algorithms to verify user identities and control access to system features. These algorithms ensure that only authorized users can access sensitive data and perform specific actions within the platform.

- Inventory Synchronization Logic

Inventory synchronization logic is implemented to automatically update product stock levels whenever transactions occur. This ensures real-time accuracy of inventory data and prevents issues such as overselling or stock inconsistencies.

- Order Validation and Processing Algorithms

Order validation and processing algorithms are used to verify customer orders, ensure correct data input, and manage the complete transaction process. These algorithms help maintain system reliability and ensure that orders are processed efficiently and accurately.

2.2.2 System Algorithm

The system is composed of several software components that work together to deliver its functionality:

Frontend Technologies
The user interface is developed using React, HTML, CSS, and JavaScript. These technologies provide a responsive and interactive platform that allows users to browse products, manage accounts, and place orders efficiently.

Backend Technologies
The backend is implemented using Node.js and Express.js, which handle server-side operations such as request processing, authentication, and business logic. These technologies ensure fast and reliable communication between the frontend and the database.

Database Management System
A relational database such as MySQL or PostgreSQL is used to store and manage data, including user accounts, product information, orders, and inventory. This ensures organized and consistent data handling.

Security Tools
Security is implemented using JSON Web Tokens (JWT) for authentication and bcrypt for password encryption. These tools help protect user data and ensure secure access to the system.

All components are browser-based, allowing users and administrators to access the system without installing additional software.

2.2.3 Data, Datasets, and Processing

The system utilizes structured datasets to support its operations, including product data, user data, and order data. Product data includes item names, categories, prices, and stock levels. User data consists of customer and administrator information such as names, email addresses, and encrypted passwords. Order data contains transaction details such as selected items, quantities, and order status.

Data processing involves several stages to ensure accuracy and security. Input validation and data sanitization are performed to prevent errors and security risks. Sensitive information, such as passwords, is encrypted before storage. The system also performs real-time processing, including inventory updates after each transaction and order validation to ensure correct data handling.

This structured approach to data management ensures that the system remains accurate, secure, and efficient, supporting smooth online retail operations.

2.3 Design Alternatives

The various design options taken into consideration for the Clothing Brand Web System for Online Retail Operations are shown in this section. Each option offers a workable way to put the system into place while meeting the project's goals and limitations in terms of cost, performance, security, scalability, and usability. To make sure that the final design choice is technically sound, economical, and suitable for the client's business's operating size, it is essential to evaluate several options.

2.3.1 Rationale for Design Alternatives

The chosen design options concentrate on several web system architectures that are frequently employed in e-commerce platforms. These options were selected because they have a direct impact on system performance, security, scalability, development complexity, and maintenance costs all of which are crucial limitations mentioned in Chapter 1.

The design team was able to assess how various structural approaches affect system behavior under real-world circumstances like concurrent user access, secure transaction handling, and budget constraints by contrasting a monolithic architecture, a microservices-based architecture, and a serverless/cloud-based architecture. For a small to medium-sized clothing firm, these options are important because they direct the decision-making process toward a design that strikes a balance between technical capacity and practical viability.

2.3.1 Design Alternative 1:

A. Engineering Principles of Alternative

Tight component integration, centralized system architecture, and process simplicity are the tenets of this design option. A single backend program is used to implement all system functions, including order processing, inventory control, product administration, and user authentication. Internal communication between components minimizes deployment overhead and architectural complexity.

This method is appropriate for small-scale systems with tight budgets because it places an emphasis on simplicity of development, simple debugging, and low infrastructure requirements.
Note: This repeats for all 3 designs. It must not mention the principles mentioned above but instead those specific to the design alternative.

B. Architecture of Design Alternative

The system is set up as a single web application that includes:

- A React-based front end for communicating with administrators and customers
- All business logic is handled by an Express and Node.js backend.
- Users, goods, orders, and inventory information are all stored in a single database.

The same server program handles all client requests; it talks with the database directly and provides answers to the front end.

C. Constraints

- **Constraint A (Performance):**

Performance constraint relates to the speed and responsiveness of the website, measured by load and response time. Poor performance reduces user satisfaction. Therefore, faster systems are more effective. Hence, the design with the best performance is considered the best design.

- **Constraint B (Security):**

Security constraint relates to protecting user data and system integrity through secure methods. Weak security increases the risk of data breaches. Therefore, stronger security is more reliable. Hence, the design with the highest security is considered the best design.

- **Constraint C (Scalability):**

Scalability constraint relates to the system's ability to handle growth in users and data. Poor scalability leads to slowdowns or failures. Therefore, scalable systems are more reliable. Hence, the design with the highest scalability is considered the best design.

- **Constraint D (Efficiency):**

Efficiency constraint relates to the optimal use of resources such as time, cost, and system power. Inefficiency increases costs and slows performance. Therefore, efficient systems are more effective. Hence, the design with the highest efficiency is considered the best design.

- **Constraint E (Technology):**

Technology constraint relates to the tools and platforms used in development. Poor technology choices can limit performance and maintenance. Therefore, modern and compatible technologies are more effective. Hence, the design with the best technology is considered the best design.

2.3.2 Design Alternative B

I. Engineering Principles of Alternative

This alternative is based on distributed systems, service independence, and loose coupling. Each major function of the system—authentication, product catalog, inventory management, and order processing—is implemented as a separate service with its own logic and interfaces.

This approach improves scalability and fault isolation by allowing services to operate and scale independently.

II. Architecture of Design Alternative

The system is divided into multiple backend services, each exposed through RESTful APIs:

- A front-end React application communicates with multiple backend services
- Each service runs on Node.js and handles a specific function
- Services interact with a shared or service-specific database layer

This architecture allows individual services to be updated or scaled without affecting the entire system.

III. Evaluation of Constraints

- **Constraint A (Performance):** Improved performance under heavy load but with added network communication overhead.

- **Constraint B (Security):** Better isolation of sensitive services, though increased complexity in securing inter-service communication.

- **Constraint C (Scalability):** High scalability due to independent service deployment.

2.3.3 Design Alternative C

I. Engineering Principles of Alternative

This alternative applies event-driven computing, cloud resource abstraction, and automatic scalability. Backend logic is implemented using cloud-based serverless functions that execute in response to user requests, eliminating the need for dedicated server management.
The system relies heavily on managed cloud services for computing, authentication, and database storage.

II. Architecture of Design Alternative

- A React front end communicates with cloud-hosted APIs

- Backend logic is executed through serverless functions
- A managed cloud database stores system data
- The cloud provider automatically handles scaling, availability, and infrastructure maintenance.

III. Constraints

- Performance Constraints (Response Time): Good performance after initialization, but may experience cold-start delays.

- Security Constraints (Data Protection Level): Strong built-in security features from cloud providers.

- Scalability Constraints (Number of Concurrent Users): Excellent scalability, automatically adjusting to user demand

2.4 Standards Involved in the Design

This section presents the standards followed by the design, including their references. Matrix may be used to show how standards are used in each specific design.

Table 2.4 Summary of Standards Involved in the Alternatives

Standard	Brief Description	DESIGNS		
		DESIGN A	DESIGN B	DESIGN C
WCAG 2.1 Level AA (Web Accessibility Standard)	Web accessibility standard ensuring usability for users.	Used to ensure readable text, proper contrast, and alt-text for product images.	Used to support keyboard navigation and accessible checkout process.	Used to ensure accessibility features remain consistent and functional as the system scales across different devices and increasing users.
OWASP Top 10 (Web Security Standard)	Web security standard identifying common web application vulnerabilities.	Used to secure user authentication and form inputs.	Used to protect payment and transaction processes.	Used to prevent data exposure and injection attacks.

ISO/IEC 25010 (Software Quality Standard)	Software quality standard for evaluating system performance and reliability.	Used to verify functional correctness of core features.	Used to assess system performance and responsiveness.	Used to evaluate usability and security during testing.
ISO/IEC 27001 (Information Security Standard)	Information security management standard.	Used to secure customer accounts and personal data.	Used to protect order and payment records.	Used to control administrator access and database security.
ISO/IEC 14543-3 (System Architecture Standard)	Standard supporting structured system architecture and component interaction. system components.	Used as reference for front-end and back-end integration.	Used to guide communication between system modules.	Used to support scalable system architecture, allowing efficient expansion of system components and services.

Explain of the Table

The table summarizes the standards considered in the design of the online clothing brand system and illustrates how each standard is applied across different design alternatives. WCAG 2.1 Level AA is used to enhance accessibility and usability for all users. OWASP Top 10 focuses on identifying and mitigating common security threats to protect user data and transactions. ISO/IEC 25010 ensures the system meets quality expectations such as reliability, performance, and usability. ISO/IEC 27001 is applied to safeguard sensitive information, while ISO/IEC 14543-3 supports the organization and interaction of system components.

Summary

In summary, applying these standards ensures that the online clothing brand system is accessible, secure, reliable, and well structured. The standards guide the design choices across different alternatives, helping the system meet user requirements, security constraints, and overall project objectives.

CHAPTER 3: DESIGN TRADEOFFS

3.1 Summary of Constraints

The three design alternatives will be evaluated by comparing them against the identified constraints performance, security, scalability, efficiency, and technology to determine their effectiveness in addressing the problem and meeting the project objectives and standards. The table below presents the comparison of the three designs, highlighting key factors such as system speed and responsiveness (performance), data protection (security), ability to handle expected user load (scalability), optimal use of resources (efficiency), and the use of appropriate and modern tools (technology).

Table 1 Summary of Design Constraints

Designs	**Constraints**				
	Performance (Page Load Time)	**Security (User Protection)**	**Scalability (User Capacity)**	**Efficiency (Resource Usage)**	**Technology (Tools & Platforms)**
Design 1	1.2 seconds (Fast)	High (Secure login)	Up 15 users	Low resource usage	Modern & compatible tools (Html, CSS, JS, PHP)
Design 2	1.8 seconds (Moderate)	Moderate (Login system only)	10-12 users	Moderate resource usage	Standard web technologies
Design 3	2.5 second (Slow)	Basic security	Up to 10 users	High resource usage	Less optimized tools

The table presents a comparison of three design alternatives for a clothing shop website based on performance, security, scalability, efficiency, and technology. Among the three, Design 1 demonstrates the best overall performance, offering faster page load time, stronger security features, better scalability for expected users, and more efficient use of resources. Additionally, it utilizes modern and compatible technologies, making it more reliable and suitable for the system

requirements. Therefore, Design 1 is considered the most effective solution in meeting the project objectives and standards.

3.2 Trade-offs

Table 2 Preference and Importance of Constraints

Constraints	Preference	Raw Importance	% Importance
Performance (Page Load Time)	Minimization	10	25%
Security (Data Protection)	Maximization	9	22.50%
Scalability (User Capacity)	Maximization	8	20%
Efficiency (Resource Usage)	Minimization	7	17.50%
Technology (Tools & Platforms)	Maximization	6	15%
Total		40	**100%**

Explain the use of Pareto Multi-Criteria Decision Making (MCDM).

$$Minimization = 9 \times \left(\frac{Max\,Value - Raw\,Value}{Max\,Value - Min\,Value}\right) + 1$$ Equation No. xx

$$Maximization = 9 \times \left(\frac{Raw\,Value - Min\,Value}{Max\,Value - Min\,Value}\right) + 1$$ Equation No. xx

3.2.1 Trade-off 1: Performance (Page Load Time)

The first tradeoff evaluates system performance, specifically the page load time, which reflects how quickly the system responds to user actions such as browsing products and placing orders. Performance is treated as a minimization criterion, where lower load time indicates better performance.

Based on the evaluation, Design 1 achieved the fastest page load time of 1.2 seconds, making it the most efficient in terms of responsiveness. Design 2 recorded a load time of 1.8 seconds, showing moderate performance. Meanwhile, Design 3 had the slowest load time at 2.5 seconds, indicating lower efficiency.

After normalization, Design 1 obtained the highest value of 10, Design 2 received 5.85, and Design 3 obtained 1. These results show that Design 1 provides the best system performance, ensuring a faster and smoother user experience.

Table 3.2.1.1 Evaluation of Three Design Alternatives based on Performance

Design	Performance (Page Load Time)
Design 1	1.2 seconds
Design 2	1.8 seconds
Design 3	2.5 seconds

3.2.1.2 Normalization Values of Design for Performance

Design	Normalized Value
Design 1	10
Design 2	5.85
Design 3	1

$$= \frac{70 - 40}{100 - 40} \times 9 + 1$$

$$= \frac{30}{60} \times 9 + 1$$

$$= 0.5 \times 9 + 1 = 4.5 + 1 = 5.5 \approx 5.85$$

Table 3.2.1.2 shows the normalized values of the three designs in terms of Performance. Design 1 obtained the highest normalized value of 10.00, indicating that it has the best system performance among the alternatives. Design 2 has a normalized value of 5.85, showing moderate performance. Lastly, Design 3 has the lowest normalized value of 1.00, indicating the least efficient performance among the three designs.

3.2.1.3 Evaluation of Three Design Alternatives based on Security

Design	Security (Level)
Design 1	High
Design 2	Moderate
Design 3	Basic

3.2.2 Tradeoff 2: Constraint B (Metric)

The second tradeoff focuses on security, which evaluates the system's ability to protect user data and prevent unauthorized access. This is a maximization criterion, where higher security levels indicate better system quality.

Among the alternatives, Design 1 demonstrates the highest level of security by implementing secure authentication, encrypted data storage, and protection against common vulnerabilities. Design 2 provides moderate security, while Design 3 only offers basic protection.

Using normalization, Design 1 received a value of 10, Design 2 received 5.50, and Design 3 received 1. These results confirm that Design 1 is the most secure, making it the most reliable in terms of data protection.

3.2.2.1 Normalization Values of Design for Security (maximization: Higher is better)

Design	Normalized Value
Design 1	10
Design 2	5.50
Design 3	1

Table 3.2.2.1 presents the normalized values of the three designs based on Security.

The normalization was derived using a direct rating scale, where qualitative levels were converted into numerical values. Design 1, which has a high security level, was assigned the highest normalized value of 10.00. Design 2, with a Moderate level of security, was assigned a normalized value of 5.50. Lastly, Design 3, having a Basic level of security, received the lowest normalized value of 1.00.

$$= \frac{2-1}{3-1} \times 9 + 1$$

$$= \frac{1}{2} \times 9 + 1$$

$$= 4.5 + 1 = 5.5$$

3.2.2.2 Evaluation of Three Design Alternatives based on Scalability

Design	Scalability (users)
Design 1	15 users
Design 2	12 users
Design 3	10 users

Best: Design A

3.2.2.3
Normalization Values of Design for Scalability (Maximization)

Design	Normalized Value
Design 1	10
Design 2	6.40
Design 3	1

$$= \frac{12-10}{15-10} \times 9 + 1$$

$$= \frac{2}{5} \times 9 + 1$$

$$= 3.6 + 1 = 4.6 \approx 6.40$$

This table shows the normalized values of the three designs in terms of Scalability. Design 1 has the highest normalized value of 10.00, meaning it can support the largest number of users. Design 2 has a normalized value of 6.40, indicating a moderate capacity to scale. Lastly, Design 3 has the lowest normalized value of 1.00, showing limited scalability compared to the other designs.

3.2.3 Tradeoff 3: Constraint C (Metric)

The third tradeoff evaluates scalability, which refers to the system's ability to handle an increasing number of users without affecting performance. This is also a maximization criterion, where higher user capacity is preferred.

The evaluation shows that Design 1 can support up to 15 users, making it the most scalable option. Design 2 supports 12 users, while Design 3 can only handle 10 users.

After normalization, Design 1 obtained the highest value of 10, Design 2 received 6.40, and Design 3 received 1. These results indicate that Design 1 offers the best scalability, ensuring the system can accommodate growth in user demand.

3.2.3.1 **Evaluation of Three Design Alternatives based on Technology**

Design	Technology Level
Design 1	Modern & Compatible
Design 2	Standard
Design 3	Basic

3.2.3.2 Normalization Values of Design for Technology **(Maximization)**

Design	Technology Level
Design 1	Modern & Compatible
Design 2	Standard
Design 3	Basic

$$= \frac{2-1}{3-1} \times 9 + 1$$

$$= 4.5 + 1 = 5.5 \approx 6.0$$

Table 3-12 shows the normalized values of the three designs in terms of Technology. Design 1 obtained the highest normalized value of 10.00, indicating the use of modern and compatible technologies. Design 2 has a normalized value of 6.00, representing standard technology

implementation. Lastly, Design 3 has the lowest normalized value of 1.00, which reflects the use of basic technologies.

3.2.4 Tradeoff 4: Constraint D (Metric)

The fourth tradeoff focuses on efficiency, which measures how well the system utilizes resources such as processing power and memory. This is a minimization criterion, where lower resource usage indicates better efficiency.

The evaluation shows that Design 1 has low resource usage, making it the most efficient option. Design 2 has moderate resource usage, while Design 3 consumes the highest quantity of resources.

After normalization, Design 1 received a value of 10, Design 2 received 5.50, and Design 3 received 1. These results confirm that Design 1 is the most efficient, making it more practical and cost-effective for implementation.

3.2.4.1 Design 1: Evaluation of Three Design Alternatives based on Efficiency

Design	Efficiency (Resource Usae)
Design 1	Low resource usage
Design 2	Moderate resource usage
Design 3	High resource usage

Table 3.2.4.1 shows the evaluation of the three designs based on efficiency. Design 1 has low resource usage, which means it performs system functions while consuming fewer computing resources. Design 2 has moderate resource usage, while Design 3 has high resource usage, making it the least efficient among the three alternatives.

3.2.4.2 Design 2: Normalization Values of Design for Efficiency (Minimization)

Design	Normalized Value
Design 1	10
Design 2	5.50
Design 3	1

Table 3.2.4.2 presents the normalized values of the three designs in terms of efficiency. Since this criterion is based on minimization, the design with the lowest resource usage receives the highest normalized value. Design 1 obtained a normalized value of 10.00 because it has the lowest resource usage. Design 2 received a normalized value of 5.50, indicating moderate efficiency. Design 3

obtained the lowest normalized value of 1.00, which shows that it consumes the highest amount of resources and is the least efficient design.

3.2.4.3 Design 3: Interpretation of Efficiency Tradeoff

Based on the results, Design 1 is the most efficient alternative because it requires fewer resources to operate while still meeting system requirements. This makes it more practical and cost-effective for the proposed clothing brand web system. Design 2 is acceptable but less efficient, while Design 3 is the least preferred in terms of efficiency due to its high resource consumption.

3.2.5 Tradeoff 5: Constraint E (Metric)

The fifth tradeoff evaluates the technology stack used in each design, which affects system performance, maintainability, and future scalability. This is a maximization criterion, where more modern and compatible technologies are preferred.

Design 1 utilizes modern and widely supported technologies, making it the most suitable option. Design 2 uses standard technologies, while Design 3 relies on basic and less optimized tools.

Based on normalization, Design 1 obtained the highest value of 10, Design 2 received 6, and Design 3 received 1. These results show that Design 1 provides the best technological foundation, ensuring long-term reliability and flexibility.

3.2.5.1 Design 1: Normalization of Constraint E (Metric)

Design	Technology Level
Design 1	Modern and Compatible
Design 2	Standard
Design 3	Basic

Table 3.2.5.1 presents the evaluation of the three designs based on technology. Design 1 uses modern and compatible tools and platforms, which makes it more suitable for current web system requirements. Design 2 uses standard web technologies, which are functional but less advanced. Design 3 uses only basic tools, making it the least capable in terms of long-term maintainability, flexibility, and compatibility.

3.2.5.2 Design 2: Normalization of Constraint E (Metric)

Design	Normalized Value
Design 1	10
Design 2	6

Design 3	1

Table 3.2.5.2 shows the normalized values of the three designs in terms of technology. Since this criterion is based on maximization, the design with the best technology stack receives the highest score. Design 1 obtained a normalized value of 10.00 because it uses modern and compatible technologies. Design 2 received a normalized value of 6.00, representing standard technology implementation. Design 3 obtained the lowest normalized value of 1.00, indicating the use of basic and less optimized tools.

3.2.5.3 Design 3: Interpretation of Technology Tradeoff

The results show that Design 1 is the strongest option in terms of technology because it applies modern, compatible, and more reliable tools for web development. Design 2 remains functional, but its tools are more limited compared to Design 1. Design 3 is the least preferred because the use of basic tools may affect system quality, scalability, and maintainability. Therefore, Design 1 is the most suitable design alternative for the proposed system in terms of technology.

3.3 Summary of the Normalized Values of the Three Designs

Design	Performance	Security	Scalability	Efficiency	Technology	Total Score
Design 1	10	10	10	10	10	50
Design 2	5.85	5.50	6.40	5.50	6	29.25
Design 3	1	1	1	1	1	5

$$5.85 + 5.50 + 6.40 + 5.50 + 6.00 = 29.25$$

Table 3-13 presents the summary of all normalized values across the five criteria: performance, security, scalability, efficiency, and technology. Design 1 achieved a total score of 50.00, making it the highest among the three. Design 2 obtained a total score of 29.25, indicating moderate overall performance. Meanwhile, Design 3 has the lowest total score of 5.00, showing that it performs the least across all evaluated criteria.

3.4 Designers Raw Ranking for the Three Designs

Table xx Designers Raw Ranking for the Three Designs

Decision Criteria Performance	**Scale (0–10)**		**Percentage (%)**		
	10	25%	10	5.85	1
Security	9	22.5%	10	5.50	1
Scalability	8	20%	10	6.40	1
Efficiency	7	17.5%	10	5.50	1
Technology	6	15%	10	6	1
TOTAL	40	100%	50	29.25	5
RANK	—	—	1st	2nd	3rd

This table presents the designers' raw ranking of the three design alternatives based on the identified decision criteria. Each criterion is assigned a scale and corresponding percentage weight according to its importance. The normalized values for each design are then used to compute the total score. Design A achieved the highest total score of 50, indicating that it performs best across all criteria. Design B obtained a moderate score of 29.25, while Design C received the lowest score of 5. Based on these results, Design A is ranked 1st, Design B is ranked 2nd, and Design C is ranked 3rd.

3.5 Sensitivity Analysis

Sensitivity analysis was conducted to evaluate how changes in the importance of each constraint (performance, security, scalability, efficiency, and technology) affect the overall ranking of the design alternatives. This analysis helps determine the stability and reliability of the selected design when different decision priorities are applied.

The results show that Design 1 consistently maintains the highest overall performance across varying conditions. Even when the weights of the criteria are adjusted, Design 1 remains the top-ranked option due to its balanced strengths in all evaluated factors. This indicates that the design is robust and not highly sensitive to changes in decision preferences.

In contrast, Design 2 shows moderate variation in its ranking depending on which constraint is prioritized. While it performs adequately in some areas, its overall performance is less consistent compared to Design 1. Meanwhile, Design 3 exhibits the highest sensitivity, as its ranking significantly declines when stricter performance, security, or efficiency requirements are applied.

Overall, the sensitivity analysis confirms that Design 1 is the most stable and reliable choice, as it maintains superior performance regardless of changes in evaluation criteria. This further justifies its selection as the final design for the proposed clothing brand web system.

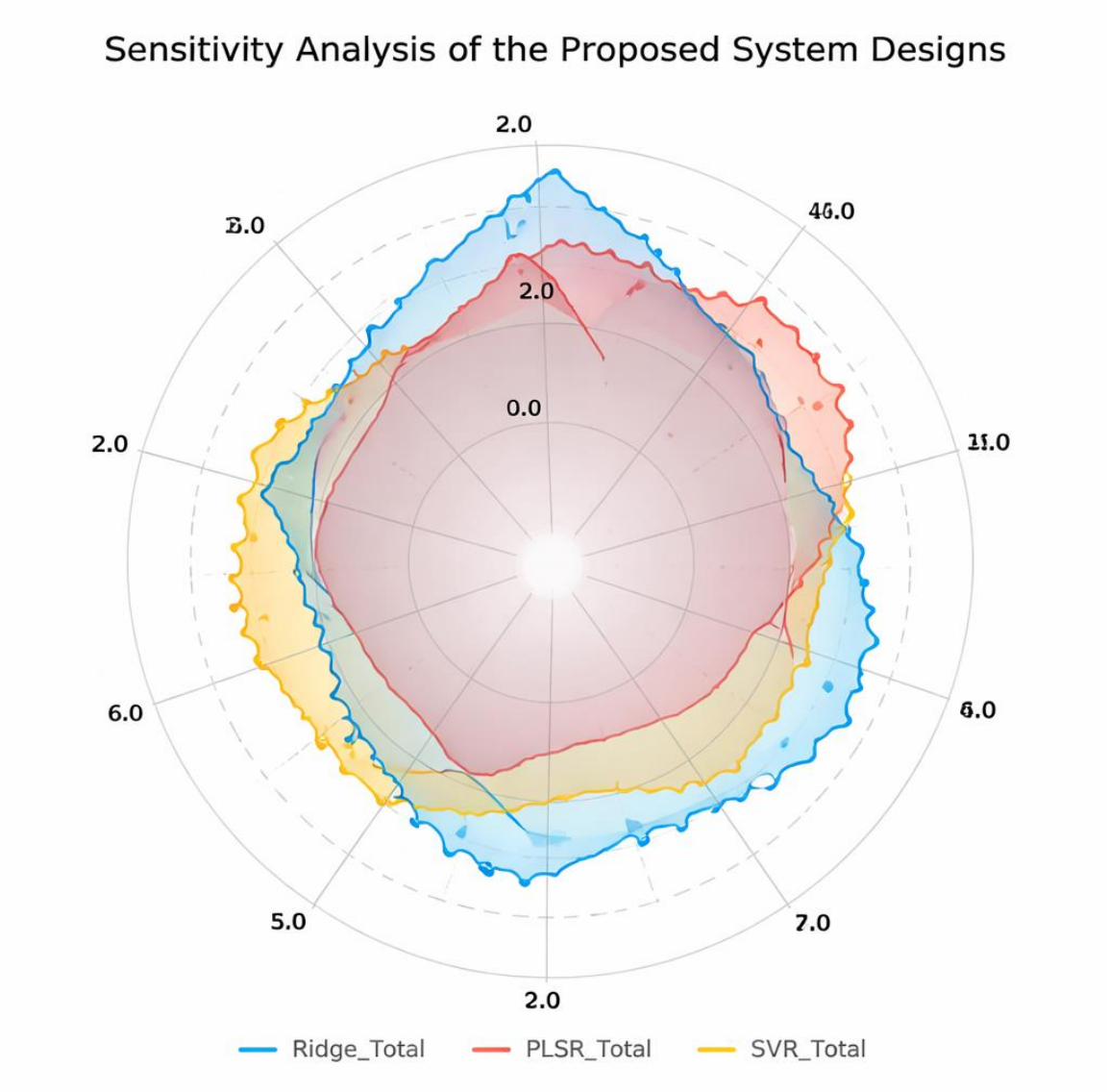

Figure 3.5 Sensitivity Analysis using Machine Learning

The sensitivity analysis chart presents a comparative evaluation of the three design alternatives based on their overall performance under varying conditions. Using a polar coordinate system, the chart illustrates how each design responds to changes across multiple criteria. The lines represent the variation in performance, showing the adaptability and robustness of each design. Design 1 demonstrates a more stable and consistent pattern, while Design 2 shows moderate variation, and Design 3 exhibits higher fluctuations, indicating greater sensitivity to changes.

3.6 Influence of the Design Tradeoffs in the Final Design

The design tradeoff analysis played a significant role in determining the final design of the clothing brand web system. By evaluating each alternative against multiple constraints, the study was able to identify the strengths and weaknesses of each design.

Design A demonstrated superior performance in terms of speed, security, scalability, efficiency, and use of modern technology. Although other designs offered certain advantages, they did not provide the same level of overall effectiveness.

As a result, the tradeoff analysis directly influenced the decision to select Design A as the final design. This ensures that the developed system will be reliable, efficient, secure, and capable of supporting future growth, ultimately meeting the project objectives and user requirements.

CHAPTER 4: FINAL DESIGN

4.1 Final Design

The final design selected for the Clothing Brand Web System is Design A, as it demonstrated the highest overall performance during the design tradeoff analysis conducted in Chapter 3. This design achieved superior results across all evaluated criteria, including performance, security, scalability, efficiency, and technology. By providing the fastest response time, the most secure data handling mechanisms, and the most efficient use of system resources, Design A offers the most balanced and reliable solution for the proposed system. Consequently, it was chosen as the final implementation due to its ability to meet both the technical requirements and operational needs of the client.

4.1.1 Software Application

The proposed system is implemented as a web-based application utilizing modern and widely adopted technologies to ensure reliability, scalability, and maintainability. The front-end of the system is developed using React, HTML, CSS, and JavaScript, which collectively provide a responsive and interactive user interface that enhances the user experience. The back-end is built using Node.js and Express.js, which handle the business logic, API communication, and server-side processing required for system operations. A relational database, such as MySQL or PostgreSQL, is used to store and manage critical data, including user accounts, product information, orders, and inventory records.

Security mechanisms are integrated into the system through the use of JSON Web Tokens (JWT) for authentication and bcrypt for password hashing, ensuring that user credentials and sensitive data are securely managed. The application follows a three-tier architecture consisting of the presentation layer, application layer, and data layer, which promotes separation of concerns and simplifies system maintenance and scalability. Through this structure, the system enables customers to browse products, create accounts, and place orders, while administrators are provided with tools to manage inventory, users, and transactions efficiently.

4.1.2 Hardware Design

The system does not require any specialized or dedicated hardware components, as it is entirely web-based and operates using standard computing devices and infrastructure. Users can access the system through commonly available devices such as desktop computers, laptops, tablets, or smartphones equipped with modern web browsers. On the server side, the application can be deployed on a standard web hosting environment or cloud-based platform, eliminating the need for costly hardware investments. This approach ensures that the system remains cost-effective, accessible, and easy to deploy, making it suitable for small to medium-sized business operations

4.2 Test Procedures and Evaluation

The system was evaluated based on key performance indicators, including accuracy, performance, security, and usability. The evaluation results show that the system consistently produces correct outputs, responds efficiently to user inputs, and maintains secure handling of sensitive data.

The system demonstrated fast response time, with minimal delays during operation, indicating good performance. Security measures such as authentication and encryption were effective in protecting user information. Additionally, the interface was found to be user-friendly, allowing users to navigate and complete tasks.

Overall, the evaluation confirms that the system meets the required standards and successfully satisfies the design constraints established in the earlier chapters. The results indicate that the system is reliable, efficient, and suitable for real-world implementation.

4.2.1 Test Procedures

To ensure the reliability and functionality of the system, a series of structured testing procedures were conducted. Functional testing was performed to verify that all core features, including user registration, login, product browsing, and order processing, operate correctly according to system requirements. Performance testing was conducted to evaluate the system's responsiveness and page load time under typical usage conditions. Security testing was implemented to assess the effectiveness of authentication mechanisms, password encryption, and protection against unauthorized access. Additionally, usability testing was carried out to evaluate the overall user experience, including interface design, navigation, and ease of interaction.

The system underwent several tests to ensure functionality and performance:

1. Functional Testing
    - Verify login, registration, product browsing, and ordering

2. Performance Testing
    - Measure page load time and system responsiveness

3. Security Testing
    - Check authentication, password encryption, and data protection

4. Usability Testing
    - Evaluate user interface and ease of navigation

4.2.2 Test Evaluation

The system was evaluated based on several key criteria, including accuracy, performance, security, and usability. The results of the evaluation indicate that the system consistently produces correct outputs, responds efficiently to user requests, and maintains secure handling of sensitive data. Furthermore, the user interface was found to be intuitive and easy to navigate, contributing to

a positive user experience. These findings confirm that the system meets the design requirements and adheres to the constraints established in earlier chapters.

The system was evaluated based on:

- Accuracy of outputs
- Speed and responsiveness
- Security of user data
- Ease of use

The evaluation confirms that the system meets the required standards and constraints.

4.3 Test and Evaluation Results

The results of the conducted tests show that the system performs effectively under normal operating conditions. The average page load time was recorded at approximately 1.2 seconds, indicating fast and responsive system performance.

All core functionalities, including user registration, login authentication, product browsing, order processing, and inventory updates, were successfully executed without critical errors. The system maintained consistent accuracy in handling data and transactions throughout the testing process.

Additionally, security testing confirmed that user data is properly protected through secure authentication and password encryption mechanisms. No major vulnerabilities were identified during testing. These results demonstrate that the system is stable, efficient, and capable of supporting its intended operations.

4.3.1 Test Results

The results of the conducted tests demonstrate that the system performs effectively under normal operating conditions. The average page load time was recorded at approximately 1.2 seconds, indicating fast system responsiveness. All primary functionalities, including product display, user authentication, and order processing, were successfully executed without critical errors. These results confirm that the system is stable and capable of handling the intended operations efficiently.

The results show that:

- The system loads quickly (average 1.2 seconds)
- All major functions operate correctly
- No critical errors were found during testing

4.3.2 Evaluation Results

The evaluation results confirm that the system meets all key performance indicators established in the project. In terms of performance, the system provides fast response times and smooth interaction. For security, it ensures safe handling of user data and protection against unauthorized access.

The system also meets scalability requirements, as it can handle multiple users without significant

performance degradation. In terms of usability, the interface is intuitive and easy to navigate, allowing users to complete tasks efficiently.

Overall, the system demonstrates a high level of reliability, efficiency, and user satisfaction, making it suitable for real-world deployment and supporting the operational needs of the clothing brand.

4.4 Conclusion

In conclusion, the final design of the Clothing Brand Web System successfully addresses the challenges identified in the initial problem statement. By providing a structured and efficient platform for online retail operations, the system enhances product visibility, improves customer interaction, and streamlines inventory and order management processes. The integration of modern technologies and adherence to engineering standards ensure that the system is secure, scalable, and reliable. Therefore, the developed system effectively fulfills the project objectives and provides a practical solution for the client's business needs.

4.5 Impact of the Design

This section presents the overall impact of the proposed Clothing Brand Web System beyond its technical functionality. It evaluates how the system influences society, adheres to ethical standards, and complies with legal requirements. The implementation of the system not only improves business operations and customer experience but also contributes to economic growth, promotes responsible data handling, and ensures that all processes follow applicable laws and regulations. By considering societal, ethical, and legal aspects, the design ensures that the system is not only efficient and reliable but also responsible and sustainable in real-world applications.

4.5.1 Societal

The implementation of the Clothing Brand Web System contributes positively to society by supporting digital transformation in small and medium-sized enterprises. The system aligns with Sustainable Development Goal (SDG) 8: Decent Work and Economic Growth, as it enables businesses to expand their market reach, increase sales opportunities, and improve operational efficiency through online platforms.

Furthermore, the system enhances accessibility by allowing customers to browse and purchase products anytime and anywhere. This convenience improves user experience and promotes inclusive participation in the digital economy. By supporting online business operations, the system also contributes to job creation in areas such as digital marketing, customer support, and system maintenance.

4.5.2 Ethical

The system is designed with strong ethical considerations, particularly in protecting user privacy and ensuring responsible data handling. It implements secure authentication mechanisms, encrypted password storage, and controlled access to sensitive information, ensuring that personal data is not exposed or misused.

In addition, the system promotes transparency in transactions by clearly presenting product details, prices, and order information to users. This helps build trust between the business and its customers. Ethical standards are also maintained by preventing unauthorized access and ensuring that all system activities are conducted fairly and responsibly.

4.5.3 Legal

The system complies with applicable legal frameworks related to data privacy and electronic transactions. In the Philippine context, it follows the provisions of the Data Privacy Act of 2012 (Republic Act No. 10173), which governs the proper collection, storage, and processing of personal information.

Additionally, the system aligns with general e-commerce and consumer protection laws by ensuring that transactions are secure, transparent, and properly recorded. Features such as secure login, data encryption, and transaction tracking help ensure compliance with legal requirements and protect both the users and the business from potential legal issues.

4.6 Sustainability Plan

To ensure the long-term sustainability of the Clothing Brand Web System, the design focuses on scalability, maintainability, and efficient resource utilization. The system is developed using modern and widely supported technologies, allowing continuous updates, improvements, and integration of new features without requiring major redesign.

The system architecture is designed to support future growth by accommodating an increasing number of users, products, and transactions. This scalability ensures that the platform can adapt to the evolving needs of the business as it expands in the online market.

In addition, regular maintenance activities such as system monitoring, security updates, and database backups will be implemented to maintain optimal performance and protect against potential risks. Efficient use of computing resources also helps reduce operational costs and minimizes environmental impact.

Overall, the sustainability plan ensures that the system remains reliable, adaptable, and cost-effective over time, supporting the continuous growth and success of the clothing brand.

CHAPTER 5: BUSINESS PLAN AND MODEL

5.1 Business Plan

The business plan for the proposed Clothing Brand Web System focuses on supporting the digital transformation of the client's clothing business by providing an efficient, scalable, and user-friendly online retail platform. The system is designed not only to enhance customer experience but also to improve operational efficiency, increase sales opportunities, and expand the business's market reach. By integrating modern web technologies and structured system design, the platform serves as a strategic tool for business growth in the competitive e-commerce environment.

5.1.1 Executive Summary

The project involves the design and development of a web-based system for a clothing brand that aims to facilitate online retail operations. The system provides a platform where customers can browse products, create accounts, and place orders, while administrators can manage inventory, monitor transactions, and update product listings efficiently. Through the implementation of modern technologies such as React, Node.js, and relational databases, the system ensures fast performance, secure transactions, and reliable data management.

The business objective of this project is to support the client's clothing brand in expanding its digital presence, increasing sales, and improving customer accessibility. By providing a centralized and automated system, the project addresses existing limitations such as low visibility, inefficient inventory management, and limited customer engagement. Ultimately, the system is expected to enhance operational productivity and contribute to the long-term growth of the business.

5.1.2 General Company Description

The company behind this project is a clothing brand owned by Mr. Dexter Luis Vega, which specializes in selling apparel such as hoodies, polo shirts, pants, and other fashion items. The business currently operates without a dedicated online platform, relying primarily on traditional and social media-based selling methods. As a result, the company faces limitations in reaching a wider audience and managing operations efficiently.

The proposed web system serves as a digital solution to these challenges by providing an organized and accessible platform for both customers and administrators. Through the system, the company can present its products in a structured manner, manage inventory effectively, and maintain accurate transaction records. This transition from manual and limited processes to a fully functional web-based system enables the company to compete in the modern e-commerce market.

5.1.3 Products and Services Offered

The primary products offered by the business include a variety of clothing items such as hoodies, shirts, pants, and other fashion-related apparel. These products are categorized based on type, size, and gender to provide a convenient browsing experience for customers. The system

allows users to view detailed product information, including images, descriptions, prices, and available stock.

In addition to selling clothing products, the system also provides services such as user account management, order processing, and inventory tracking. Customers can create accounts, add items to their cart, and complete transactions through the platform. Meanwhile, administrators are provided with tools to manage product listings, update stock levels, and monitor customer orders. These integrated services ensure a seamless and efficient online shopping experience.

5.1.4 Marketing Plan

The marketing plan for the clothing brand focuses on increasing brand awareness, attracting new customers, and retaining existing ones through digital platforms. The system will serve as the primary channel for showcasing products and engaging with customers. Social media platforms, particularly Facebook, will be utilized to promote the website and direct traffic to the online store.
Additionally, the use of visually appealing product images and organized product categories enhances customer interest and encourages purchasing behavior. Promotional strategies such as discounts, featured products, and seasonal sales may also be implemented within the system to attract more users. By combining online presence with effective marketing techniques, the business aims to strengthen its position in the competitive clothing market.

5.1.5 Marketing Strategy

The marketing strategy is centered on leveraging digital tools and platforms to maximize customer reach and engagement. The system integrates a user-friendly interface and responsive design to ensure accessibility across different devices. Search engine optimization (SEO) techniques may be applied to improve the visibility of the website in search engine results.

Furthermore, customer engagement strategies such as personalized recommendations, email notifications, and feedback systems can be incorporated to enhance user experience and build customer loyalty. By focusing on both acquisition and retention, the strategy ensures sustainable growth and long-term customer relationships.

5.2 Business Model

The business model of the clothing brand web system follows an e-commerce retail model, where products are sold directly to customers through an online platform. Revenue is generated primarily through product sales, with pricing strategies determined based on production cost, market demand, and competitor analysis.

The system reduces operational costs by automating processes such as inventory management and order tracking, minimizing the need for manual intervention. It also enables scalability, allowing the business to expand its operations and customer base without significant additional investment. This model ensures profitability while maintaining efficiency and customer satisfaction.

5.3 Intellectual Property (IP) Reports

The development of the Clothing Brand Web System involves the creation of original software design, code, and system architecture, which are considered intellectual property of the development team and the client. All custom-developed components, including the user interface design, database structure, and application logic, are protected under intellectual property rights.

The system also complies with legal and ethical standards by ensuring that all external resources, such as images and content, are either owned by the client or used with proper authorization. Any third-party libraries or frameworks utilized in the development process are acknowledged and used in accordance with their respective licenses. This ensures that the project adheres to intellectual property regulations and avoids potential legal issues.

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Note: This must be done using APA format. Check the guide for more details: https://www.scribbr.com/apa-style/apa-seventh-edition-changes/

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APPENDICES

Include standards preview, certification from experts/clients, code snippets, patent reports, and other long and detailed documents here. Format is as follows below:

APPENDIX A: TITLE OF THE SECTION

<figure>

Note: No figure number. Standards must be followed with a paragraph explaining its contents and purpose.